

8 Image Compression

- Image Compression is the art and science of reducing amount data required to represent an image

Data Redundancies

- Types of Data Redundancies:
 1. Coding Redundancy
 2. Interpixel Redundancy
 3. Psychovisual Redundancy

Coding Redundancies

- If the gray levels of an image are coded in a way that uses more code symbols than absolutely necessary to represent each gray level, the resulting image is said to contain coding redundancies
- Variable-length coding

Interpixel Redundancy

- If the value of any given pixel can be reasonably predicted from the value of its neighbors, the image is said to contain interpixel redundancies
- Run-length coding

Psychovisual Redundancy

- The eye does not respond with equal sensitivity to all information
- Certain information simply has less relative importance than other information in normal visual processing
- It can be eliminated without significantly impairing the quality of image perception
- Lossy data compression

Error-Free Compression

- Variable-Length Coding
 - Huffman Coding
 - Most popular for removing coding technique redundancy

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	
a_1	0.1	0.1	0.2	0.3	0.4
a_4	0.1	0.1			
a_3	0.06	0.1	0.1	0.1	0.1
a_5	0.04				

FIGURE 8.11
Huffman source reductions.

Huffman Coding

FIGURE 8.12
Huffman code
assignment
procedure.

Original source			Source reduction							
Sym.	Prob.	Code	1		2		3		4	
a_2	0.4	1	0.4	1	0.4	1	0.4	1	0.6	0
a_6	0.3	00	0.3	00	0.3	00	0.3	00	0.4	1
a_1	0.1	011	0.1	011	0.2	010	0.3	01		
a_4	0.1	0100	0.1	0100	0.1	011				
a_3	0.06	01010	0.1	0101						
a_5	0.04	01011								

LZW Coding

- LZW compression has been integrated into a variety of mainstream imaging file formats, including GIF, TIFF, and PDF
- The PNG format was created to get around LZW licensing requirements

LZW Coding

39	39	126	126
39	39	126	126
39	39	126	126
39	39	126	126

Dictionary Location	Entry
0	0
1	1
⋮	⋮
255	255
256	—
⋮	⋮
511	—

EXAMPLE 8.7:
LZW coding.

TABLE 8.7
LZW coding
example.

Currently Recognized Sequence	Pixel Being Processed	Encoded Output	Dictionary Location (Code Word)	Dictionary Entry
	39			
39	39	39	256	39-39
39	126	39	257	39-126
126	126	126	258	126-126
126	39	126	259	126-39
39	39			
39-39	126	256	260	39-39-126
126	126			
126-126	39	258	261	126-126-39
39	39			
39-39	126			
39-39-126	126	260	262	39-39-126-126
126	39			
126-39	39	259	263	126-39-39
39	126			
39-126	126	257	264	39-126-126
126		126		

Run Length Coding

- Images with repeating intensities along their rows (columns) can often be compressed by representing runs of identical intensities as run-length pairs, where each run-length pair specifies the start of a new intensity and the number of consecutive pixels that have that intensity
- When there are few(or no) runs of identical pixels, run-length encoding results in data expansion
- BMP file format uses a form of run-length encoding

Types of File Formats

- Popular file formats are listed below:
 - 1) GIF (Graphics Interchange Format)
 - 2) JPEG (Joint Photographic Experts Group)
 - 3) PNG (Portable Network Group)
 - 4) DICOM (Digital Imaging and COMmunication)
 - 5) SVG (Vector Graphics file format)
 - 6) TIFF (Tagged Image File Format)